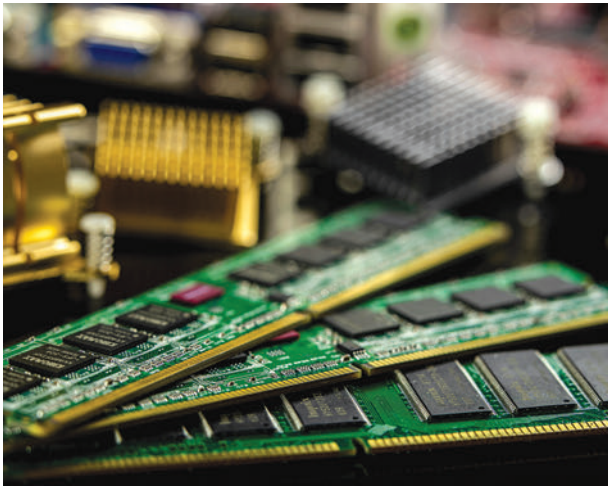




Why DDR5 replaces DDR4

DDR5 is the latest memory standard and supersedes DDR4. The transition isn't just about clock speed; DDR5 introduces a new architecture to keep pace with multi-core CPUs.

- **Two independent 32-bit channels per DIMM:** Instead of DDR4's single 64-bit channel, each DDR5 module is split into two 32-bit sub-channels, allowing the memory controller to interleave commands and reduce latency during bank conflicts. Dual-rank modules (with more memory

chips) offer four 32-bit sub-channels, boosting parallelism and throughput.

- **Increased bank groups and burst length:** DDR5 doubles the number of memory banks (32 banks within eight groups) and lengthens the burst length to 16. These changes raise the number of requests that can be processed simultaneously, delivering much higher bandwidth.
- **On-die ECC and SPD hub:** Error-correcting code (ECC) circuitry built into the memory chip corrects single-bit errors before they reach the memory controller, increasing reliability. A new SPD hub centralises the Serial Presence Detect and power management functions on the module.
- **Power management on the DIMM:** The power-management IC (PMIC), previously on the motherboard, now resides on the DIMM for DDR5. This shortens the power delivery path and allows modules to regulate voltage more precisely.
- **Higher official speeds and efficiency:** JEDEC standard DDR5 starts at 4800 MT/s and scales well beyond 8800 MT/s. The modules operate at just 1.1 V—lower than DDR4's 1.2 V—reducing power draw and heat.

Decoding RAM specs: Speed and Latency

When browsing memory kits, you'll see specifications like "32 GB (2×16 GB) DDR5-6000 CL30." Understanding each part helps you choose wisely:

PC VS LAPTOP RAM: WHAT'S THE DIFFERENCE?			
Feature	Desktop RAM (UDIMM)	Laptop RAM (SO-DIMM / LPDDR)	What It Means for You
Form Factor	Standard UDIMM (Longer) sticks, ~133mm length	Compact SO-DIMM (Shorter) sticks, ~67mm length	Desktop RAM doesn't fit in laptops and vice versa.
Installation	User-replaceable, easy to upgrade	Often soldered in modern ultrabooks	Check if your laptop RAM is upgradable before buying.
Performance (2025)	Typically higher frequencies (5600–8400 MHz) and better thermals	Slightly lower speeds (4800–6400 MHz), limited cooling	Desktops allow more aggressive tuning and XMP/EXPO support.
Voltage	Standard 1.1–1.35V for DDR5 UDIMMs	Lower (1.05–1.1V) for energy efficiency	Laptops prioritise battery life over raw performance.
Capacity Options	16–128 GB (2–4 slots common)	8–64 GB (1–2 slots common; some soldered)	Desktops offer easier expansion and higher total memory.
Cooling	Heatsinks or RGB modules for heat dissipation	Usually bare chips or thin shields	Laptop RAM runs hotter in compact spaces.
Profiles (XMP/EXPO)	Widely supported on gaming motherboards	Limited or unavailable on laptops	You can overclock RAM on desktop boards, not laptops.
Availability / Cost	Wide range of options and aesthetics	Fewer upgrade kits, higher price per GB	Desktop kits are cheaper and easier to find.
Use Case	Custom-built gaming, creator, or workstation PCs	Portable systems, productivity notebooks, ultrabooks	Choose SO-DIMM only if upgrading an upgradable laptop.